Proposal to Replace Conventional Exit Signs with Self-Illuminating Exit Signs

Executive Summary

This proposal recommends replacing the organization's existing electrically powered exit signs with self-illuminating exit signs. The proposed upgrade will reduce energy consumption, lower maintenance costs, improve emergency preparedness, and provide long-term financial savings. Traditional exit signs require continuous electrical power and regular bulb replacement, resulting in ongoing operating expenses. Self-illuminating exit signs eliminate these costs while providing reliable performance for up to twelve years. By implementing this upgrade, the organization can improve safety, increase operational efficiency, and reduce overall facility expenses.

Introduction

Exit signs are a critical component of any building's safety system. During emergencies, they help occupants quickly identify evacuation routes and exit the building safely. Because exit signs operate continuously, they consume electricity around the clock and require routine maintenance to ensure proper functionality. While conventional exit signs have been widely used for many years, they create recurring expenses related to energy consumption, bulb replacement, inspections, and repairs.

As organizations continue looking for opportunities to reduce operating costs and improve efficiency, self-illuminating exit signs provide a practical alternative. These signs require little to no maintenance, remain visible during power outages, and offer significant cost savings throughout their lifespan. Replacing existing exit signs with self-illuminating models can help the organization improve safety while reducing long-term expenses.

Problem Statement

The organization currently relies on conventional exit signs that consume electricity twenty-four hours a day, seven days a week. Although each sign uses a relatively small amount of power, the combined energy usage across a facility contributes to substantial annual utility costs. In addition, conventional exit signs require regular inspections, bulb replacements, and maintenance activities that increase labor and material expenses.

Another concern is reliability during power failures. Traditional exit signs depend on electrical systems and backup power sources that require ongoing testing and maintenance. These requirements create additional costs while increasing the possibility of equipment failure. A more efficient and dependable solution is needed to reduce expenses and improve system reliability.

Proposed Solution

This proposal recommends replacing all existing electrically powered exit signs with self-illuminating exit signs. These signs operate without electricity and can remain effective for approximately ten to twelve years with minimal maintenance. Because they do not require bulbs or dedicated electrical circuits, they eliminate many of the costs associated with traditional exit-sign systems.

A similar project completed by the Marine Corps Development and Education Center in Quantico, Virginia demonstrated the effectiveness of this solution. The project generated significant first-year savings and achieved a payback period of approximately 2.6 years. These results show that self-illuminating exit signs can provide both financial and operational benefits.

Benefits

The primary advantage of self-illuminating exit signs is cost savings. Eliminating electrical consumption reduces utility expenses, while the removal of routine maintenance requirements lowers labor and material costs. Organizations also save money by eliminating the need to purchase, store, and replace bulbs throughout the life of the signs.

In addition to financial benefits, self-illuminating exit signs improve safety and reliability. Because they do not depend on electrical power, they remain visible during power outages and emergencies. The upgrade also supports environmental sustainability initiatives by reducing overall energy consumption and lowering the facility's environmental impact.

Cost Analysis

The cost of self-illuminating exit signs varies depending on the model selected. Single-face signs generally cost between $200 and $250, while double-face signs range from $450 to $500. Installation costs are relatively low because dedicated electrical circuits are not required. Although the initial investment may be higher than traditional signs, the reduction in energy and maintenance expenses allows organizations to recover costs quickly and realize long-term savings.

Implementation Plan

Implementation will begin with a facility assessment to determine the number and location of exit signs requiring replacement. After evaluating vendors and selecting an approved supplier, the organization will purchase the necessary signs and schedule installation.

Installation will be completed during non-peak operating hours to minimize disruptions. Following implementation, energy consumption and maintenance costs will be monitored to verify savings and evaluate the success of the project.

Conclusion

Replacing conventional exit signs with self-illuminating exit signs is a cost-effective investment that offers significant operational, financial, and safety benefits. The proposed solution will reduce energy consumption, eliminate most maintenance requirements, improve reliability during emergencies, and provide long-term savings. Based on the proven success of similar projects, approval of this proposal is strongly recommended.

References

Kolin, P. C. (2022). *Successful writing at work*

Marine Corps Development and Education Center. *Self-illuminating exit signs*.